Retrieval or discovery?
Minimum evidence standards for attributing scientific findings to autonomous agents

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Abstract

Large language model (LLM) agents are increasingly reported to have *discovered* scientific results rather than merely to have executed them. Such attributions are difficult to evaluate: the same artefact—a working protocol, a measured effect, a manuscript—is produced whether the agent reasoned its way to a new mechanism or recombined material already present in its training distribution, retrieved it, or was steered there by a human. We argue that the discipline currently lacks the minimum reporting and control standards that would make the distinction decidable, and we propose five—PROBE: *Provenance* disclosure, *Replication* across independent runs, *Operator-matched* controls, *Benchmark* integrity, and *Evaluation* pre-registration. We then apply the checklist to a recent, prominent claim: an LLM agent reported to have autonomously proposed and experimentally validated a previously unreported optical mechanism, an "optical bilinear interaction" analogous to the compatibility operation in Transformer attention (arXiv:2604.27092). Working entirely from the authors' own released data and analysis code, we reproduce their published accuracies to within 5×10^{-4} and then show that the benchmark supporting the computational claim does not survive elementary integrity checks: the reported cross-split generalisation is accounted for by the presence of self-paired samples, and the remaining task accuracies are inflated by near-duplicate repeats shared between training and test folds. Under ordered-pair-grouped evaluation, pair-identity prediction is not a well-defined generalisation task, category-pair classification falls below chance, and same-category classification is not significant under the exact null distribution over all 105 admissible category assignments ($p = 102/105$). In the held-out-category task, above-chance performance occurs only when self-pairs are present in both the training and the test set; removing them from either side returns accuracy to chance. We emphasise what we do *not* claim: we find no evidence of data fabrication, the hardware measurements appear sound, and the automation achievement is real. The failure is one of evidence–claim alignment, and it is the kind that PROBE is designed to catch before publication.

1 Introduction

LLM-based agents have moved, over a short period, from assisting predefined scientific workflows to being credited with scientific results in their own right. The claims made on their behalf have escalated accordingly: not that a system executed a protocol competently, but that it *discovered* something that was not previously known.

This escalation has outrun the evidence customarily supplied to support it, for a structural reason. *The artefact produced by genuine discovery and the artefact produced by retrieval-plus-execution are the same artefact.* An agent that reasons from physical first principles to a new observable; an agent that recombines two textbook techniques it has encountered thousands of times in pretraining; an agent that retrieves a published protocol; and an agent steered by its operators toward a direction they found promising—all four emit the same thing at the end. Working code, a measured effect, a coherent manuscript. Nothing in the output separates them.

For human authorship this problem is bounded, and the bounding is so thoroughly built into scientific convention that it is easy to forget it is there. A human author's prior knowledge is finite, largely declarable, and traceable through citation; the reader can ask what was known and when. For an LLM agent the prior is a training distribution that is undisclosed, effectively unbounded, and—in precisely the cases where discovery claims are most attractive—saturated with the material at issue. Attribution therefore cannot rest on the output. It has to rest on controls and disclosures that were never previously required, because they were never previously necessary.

We propose five such requirements (Sec. 2), chosen to be individually cheap and jointly close to sufficient, and then apply them to a recent and prominent claim (Secs. 3–8). The application is instructive beyond the individual case: two of the failures we find are invisible to ordinary peer review, because they live in the split logic and the label construction of a benchmark rather than in any statement the manuscript makes. They are, however, immediately visible to anyone who runs the released code with one line changed—which is the argument for requiring that the code be released, and for looking at it.

We state at the outset what this exercise is not. We allege no misconduct, we find no evidence of fabrication, and we do not dispute that the automation demonstrated in the target work is a real achievement. Section 11 makes the boundaries explicit.

2 The Probe checklist

P — Provenance disclosure. The backbone model and version, its training cut-off relative to the publication dates of all target papers, the complete initial prompt, every candidate direction generated and discarded, the criterion by which a direction was selected and by whom, every human intervention, and the full agent trace. Without these, the causal attribution "the agent discovered X" is not merely unverified but unformulated.

R — Replication. At least three independent runs from the same starting prompt, with the distribution of outcomes reported. A single trajectory from a stochastic system cannot distinguish an architectural property from a sampling event.

O — Operator-matched control. Where a physical measurement is claimed to produce information, the control must be the digital reconstruction of the *same* quantity under the *same* operator, not a different quantity. Randomising the operator, or substituting an algebraically distinct expression, tests a different hypothesis.

B — Benchmark integrity. The evaluation must possess the structure it purports to measure; train/test partitions must respect the unit of generalisation; and every reported accuracy must be compared against the best measurement-free rule available on the same partition, and against an exact or permutation null over the admissible label assignments where one can be enumerated.

E — Evaluation pre-registration. Readout parameters, feature budgets and splits must be fixed before the comparison of interest, or the search over them must be reported in full.

3 Case study: setup

The target work couples an LLM multi-agent system to a free-space optical platform and reports three studies of increasing autonomy, culminating in an open-ended investigation that identifies an "optical bilinear interaction". Two token-dependent fields E_a, E_b are coherently superposed with controlled relative phase, scattered by a fixed medium T and detected by a square-law camera; four-step phase-shifting demodulation with blank subtraction isolates

$$B_m(a,b) \;=\; E_a^{\dagger}\left(T_m^{\dagger}T_m\right)E_b, \tag{1}$$

a per-channel complex bilinear response. The computational claim is supported by an eight-token "semantic benchmark" comprising 64 ordered pairs and four tasks: pair identity (A, 64-way), same category (B, binary), category pair (C, 16-way), and cross-split same-category generalisation to a held-out category (D).

All material below derives from the authors' own deposit (DOI 10.5281/zenodo.19890402). Our pipeline mirrors `05_result4_refine_paper/scripts/analyze_advantage_boundary.py` so that any difference is attributable to a single documented modification.

3.1 Replication

We first reproduce the released numbers without modification. For the raw-input baselines—features computed directly from $E(x), E(y)$ with no optical propagation—the maximum deviation between our values and the published ones is 5×10^{-4} across all five feature families and all four tasks. The optical families reproduce likewise, and we independently recover the authors' participation-ratio effective ranks of the Complex-B feature ensemble: 3.00 at bin = 25 and 9.62 at bin = 10, against their stated $\sim$3 and $\sim$9. Nothing that follows is a pipeline discrepancy.

4 Benchmark integrity

4.1 B1: the benchmark carries no semantic structure

The eight tokens are constructed as

```
def make_token_input(tid):
    seed = 10000 + tid * 7919
    rng  = np.random.default_rng(seed)
    phases = rng.uniform(0, 2*np.pi, (6, 6))
    return np.exp(1j * phases).flatten()      # 36 complex
```

that is, eight independent unit-modulus random phase vectors. The words (*cat*, *dog*, *hammer*, ...) and the categories (*animals*, *tools*, ...) are labels attached to random vectors; no embedding enters the pipeline. Consequently the proposition "*cat* and *dog* belong to the same category" is encoded nowhere in the inputs or in the physics. The data-generating process contains no designed semantic relation from which held-out-category generalisation could be learned; any above-chance Task-D result must therefore be attributed to a finite-sample accident or to an evaluation cue, rather than to encoded semantics recovered by the measurement.

4.2 B2: Task D is accounted for by self-pairs

For each held-out category the test set contains 4 distinct positive pairs, of which 2 are self-pairs (x, x), against 24 distinct negatives. The measurement-free identity-rule baseline "same category := $(x = y)$" therefore attains balanced accuracy $\frac{1}{2}\left(\frac{2}{4} + \frac{24}{24}\right) = 0.750$. Every value obtained under the refined held-out-category Task-D protocol reproduced here lies at or below this baseline; we do not extend the comparison to values reported under the different fixed-holdout protocol of the main manuscript. Removing all self-paired samples collapses the optical feature to chance (Fig. 1), identically at both readout scales, while the per-fold values are constant to four decimals in the unablated case—the signature of a deterministic structural cue rather than a learned relation.

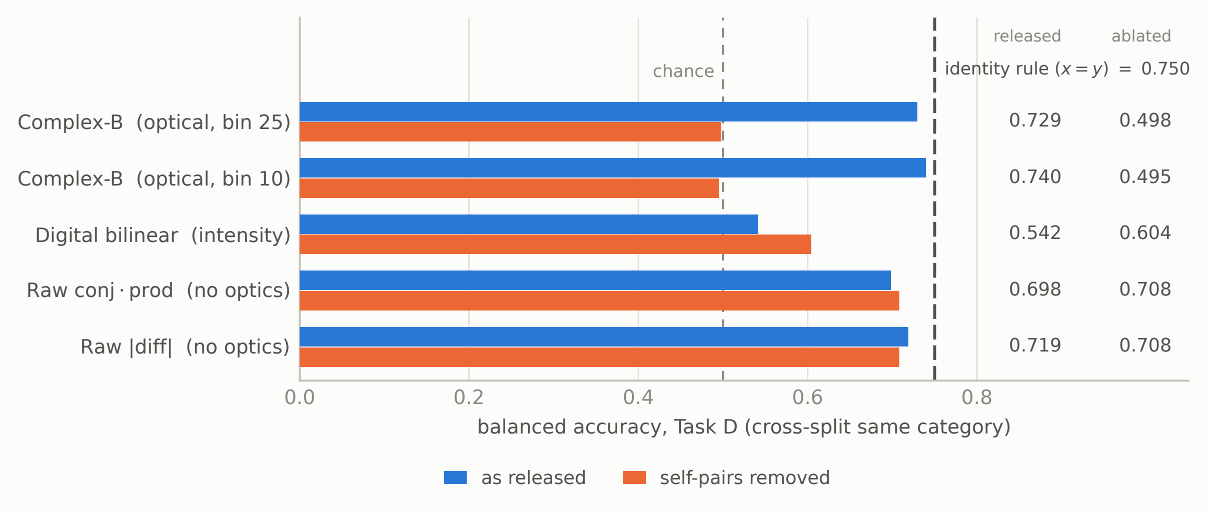

Figure 1: **Task D is accounted for by self-pairs.** Balanced accuracy on the cross-split same-category task under the refined held-out-category protocol, before and after removing every self-paired sample (x, x). The dashed rule at 0.750 is the accuracy of the measurement-free identity rule "same category := $(x = y)$", which uses the pair labels and no optical data at all. The optical Complex-B feature does not exceed that rule, and falls to chance once self-pairs are removed—identically at the coarse (bin = 25) and speckle-matched (bin = 10) readouts. The raw-input baselines, which involve no optical propagation whatsoever, are unaffected by the ablation.

Table 1: Tasks A–C at bin = 25, balanced accuracy, released scoring versus pair-grouped scoring.

Feature	Task	released	pair-grouped	Δ
Complex-B	A (64-way)	1.0000	degenerate by construction	
Complex-B	B (binary)	1.0000	0.6522	−0.3478
Complex-B	C (16-way)	1.0000	0.0474	−0.9526
Digital bilinear	B (binary)	1.0000	0.8917	−0.1083
Digital bilinear	C (16-way)	0.4437	0.1709	−0.2729

4.3 B3: Tasks A–C are inflated by repeat leakage

The released scoring applies `StratifiedKFold` over 320 samples comprising 64 pairs × 5 physical repeats, so repeats of the same pair are split between training and test folds. We measure the pairwise repeat correlation of the Complex-B feature directly: mean 0.9828, with 97.0% of repeat pairs exceeding 0.97 (Fig. 3), confirming the authors' own figure. Training and test folds thus contain near-duplicates of the same measurement.

Re-scoring with `StratifiedGroupKFold` grouped by ordered pair gives Table 1 and Fig. 2. Pair identity is not evaluable as an out-of-pair generalisation task when the ordered pair is the grouping unit, because each 64-way class corresponds to exactly one group; and Task C falls to or below the 16-class chance level of 0.0625. We verified that the Task-C collapse is not an artefact of the fold count: across $n_{\text{splits}} \in \{2, 4, 5, 8\}$ every test class is represented in training (100%) and accuracy remains in 0.043–0.109 at both readouts (Fig. 2c).

4.4 B4: the residual Task-B accuracy is not significant

Pair-grouped scoring leaves Task B at 0.6522, above the binary chance level. This residual admits an exact test. The same-category label is fixed entirely by a partition of the eight tokens into four unlabelled two-token categories, and there are $8!/((2!)^4 4!) = 105$ such partitions. We therefore

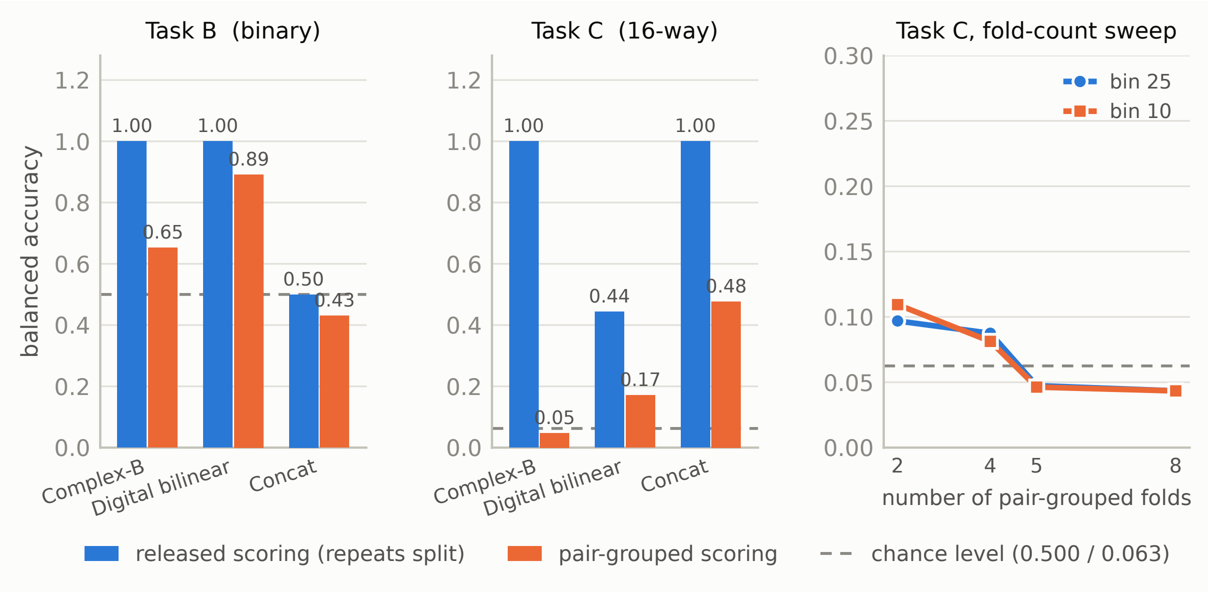

Figure 2: **Tasks B and C are inflated by repeat leakage.** (**a**,**b**) Balanced accuracy at bin = 25 under the released scoring, which splits the five physical repeats of a pair across folds, and under pair-grouped scoring, which does not. (**c**) The Task-C collapse is not a fold-count artefact: it persists across $n_{\text{splits}} \in \{2, 4, 5, 8\}$ at both readouts, with every test class represented in training in all cases.

recompute the pair-grouped Task-B accuracy under every admissible assignment. The designated semantic assignment is a member of this set, so the exact one-sided p-value is the fraction of assignments scoring at least as high as the designated one, with the designated assignment counted in the numerator; no Monte Carlo sampling and no continuity correction are involved.

The designated assignment scores 0.6522 against a null with mean 0.7280 and standard deviation 0.0347, and 102 of the 105 assignments score at least as high (Fig. 4); the exact one-sided p is 0.9714. Approximately 97% of all admissible category assignments therefore achieve accuracy at least as high as the designated semantic one. The residual is not evidence that the measurement recovers the designated category structure.

4.5 B5: the Task-D shortcut is learned, not merely present

Removing self-pairs from the training set and from the test set are distinct interventions, and crossing them separates a cue that the classifier acquires during fitting from a structural shortcut sitting in the test set. Table 2 gives the full design. Above-chance performance occurs only when self-pairs are present on *both* sides; removing them from either side returns accuracy to chance. In particular, when self-pairs are present at test time but absent during training, accuracy is 0.4917—the shortcut is available but unused. This is a learned self-pair shortcut under this evaluation protocol. We note the small effective sample: each held-out fold contributes only 4 distinct positive pairs (2 after ablation) against 24 distinct negatives.

5 Model–measurement consistency: a diagnostic observation

This section is diagnostic rather than evaluative. It bears on neither the benchmark failures of Sec. 4 nor the central attribution question, and we report it because it constrains what the released estimator can be assumed to satisfy.

Under the stated linear-optical model the unbinned channel kernel $K_p = T_p^\dagger T_p$ is *algebraically* rank-one, Hermitian and positive semidefinite; spatial binning forms $K_k = \sum_{p \in \mathcal{B}_k} K_p$, a positive

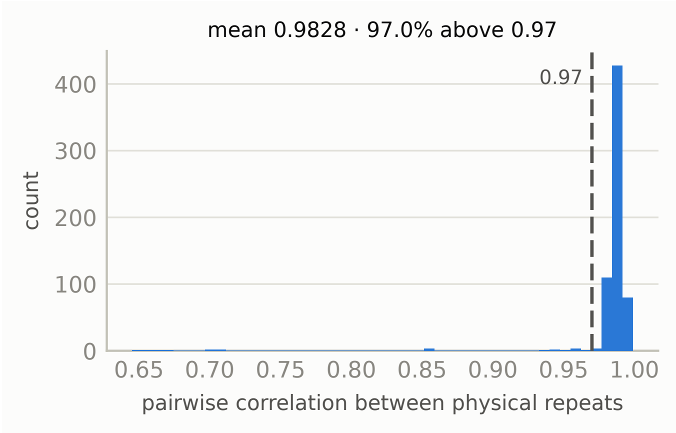

Figure 3: **Physical repeats are near-duplicates.** Distribution of pairwise correlations between the five repeated measurements of the Complex-B feature, over all 64 ordered pairs at bin = 25. Under the released scoring these near-duplicates are distributed across training and test folds.

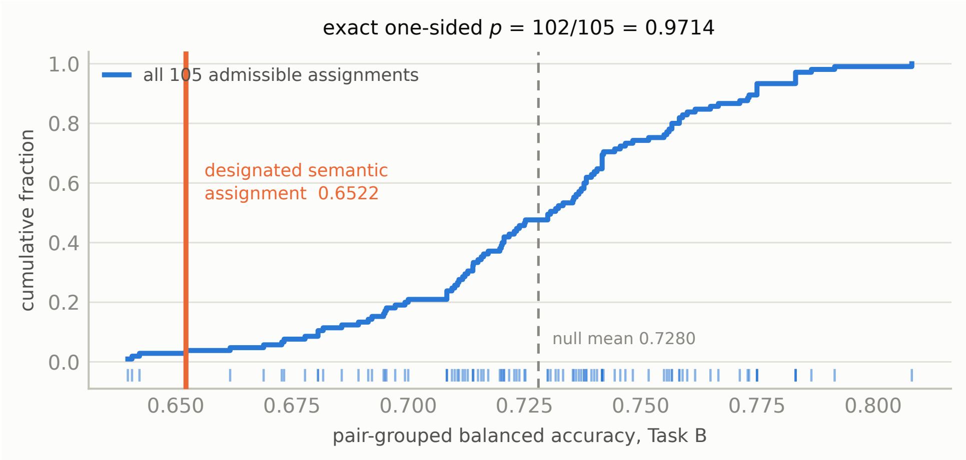

Figure 4: **The residual Task-B accuracy is not exceptional.** Empirical cumulative distribution and rug plot of pair-grouped Task-B balanced accuracy over *all* 105 partitions of the eight tokens into four unlabelled two-token categories—an exact enumeration, not a Monte Carlo sample. The designated semantic assignment lies in the lower tail.

semidefinite sum whose algebraic rank is bounded by the number of independent propagation rows in the bin. These are consequences of the model, not empirical assumptions, and they imply $B_p(y,x) = B_p(x,y)^*$.

What the release contains is an estimator carrying a reference arm, phase stepping, blank subtraction, drift and calibration error. It satisfies the ideal identities only approximately. Over the 28 off-diagonal token pairs the mean correlation between $\widehat{B}(y,x)$ and $\overline{\widehat{B}(x,y)}$ is 0.6042, against 0.1537 for the uncorrelated comparison $\widehat{B}(x,y)$; the conjugate symmetry is thus clearly present but far from exact. On self-pairs, where the model predicts a positive real value, the real part is non-negative in every channel, but the mean relative imaginary magnitude $|\mathrm{Im}\,\widehat{B}|/|\widehat{B}|$ is 0.3511.

A phase-gauge offset would reproduce this signature without implying that the cross term was not isolated, so we tested it: estimating a single global phase, and then a per-channel phase, from the self-pairs alone and applying it to all pairs. Neither accounts for the departure. The self-pair imaginary ratio falls only from 0.3511 to 0.3102 (global) and 0.3057 (per-channel), and the swap-conjugate correlation does not improve (0.5666 and 0.5606). The per-channel phases

Table 2: Task D at bin = 25 under the full 2×2 self-pair design. Identical classifier, folds and scoring throughout.

Self-pairs in training	Self-pairs in test	balanced accuracy	test +/−
present	present	0.7292	20 / 120
present	removed	0.4792	10 / 120
removed	present	0.4917	20 / 120
removed	removed	0.4979	10 / 120

are moreover nearly identical (circular spread 0.0019), so there is no channel-dependent phase structure to remove.

We draw no conclusion about the cause. Two candidate explanations remain open and are not separable from the released data: a residual in the blank or reference subtraction, and the two input arms not sharing a transmission operator, $T_x \neq T_y$. We flag that the second would also weaken one line of criticism available against the attention analogy, since a kernel $T_x^{\dagger} T_y$ is rank-one but neither Hermitian nor positive semidefinite, and is in that respect *less* constrained than the shared-operator case. Resolving this requires acquisition-level information that the deposit does not contain.

6 The missing operator-matched control

Equation (1) is the cross term of two-beam interference, isolated by four-step phase shifting. Under linear optics the physically demodulated B and the digital product $(TE_a)^* \odot (TE_b)$ formed from separately measured complex fields are the same quantity up to noise. The decisive control is therefore an equality test between them under the same T.

The released controls are not this test. The "digital bilinear" baseline is $z(x)\, z(y)$ where z is binned camera *intensity*, discarding phase; a further variant is described in the supplement as a per-channel product "no cross-conjugation"; and the pseudo-B control replaces the physical T with a random complex Gaussian. The first two compare different algebraic objects; the third asks whether a real scattering matrix differs from a random one, not whether joint measurement yields information unavailable to separate measurement. The required control is absent, and—because this experiment records only single-token *intensity*—it cannot be performed on the released data. It can, however, be performed on the same apparatus: the self-referenced transmission-matrix measurement in the earlier studies of the same work recovers exactly the complex fields it requires.

7 Evaluation search space

The evaluation supporting the computational claim involves several free choices, and the released material shows that more than one value of most of them was examined. Table 3 records the search space that is publicly visible, alongside what the public record does and does not establish about when each choice was fixed. We stress the distinction: for none of these choices do the released materials document the selection *timing*, and we therefore do not assert that any of them was made after inspecting the outcome. What the record does establish is that the reported configuration was selected from a searched space without an independent confirmation set. This is adaptive evaluation, and the standard remedy—pre-registration of the readout parameters and splits, or a held-out physical realisation on which the selected configuration is evaluated once—was not applied.

Table 3: Publicly visible evaluation search space for the Result-4 benchmark.

Choice	Publicly observed search	Selection timing	Independent confirmation
Detector bin	5, 10, 25	Not publicly documented	None identified
Output budget	multiple n_{out} values	Not publicly documented	None identified
Task-D protocol	multiple descriptions and implementations	Protocol evolution observed	None identified
Feature family	multiple optical and digital families	Exploratory analysis evident	None identified
Token vocabulary	one fixed random codebook	Not publicly documented	No independent vocabulary

8 Provenance

The central claim is causal: that an agent, rather than its training distribution or its operators, originated the mechanism. Searching the full deposit we find no backbone model name or version, no system or initial prompts, no agent trace, no record of human intervention, and no direction-selection log; the deposit states explicitly that the engine and hardware-control software are not included. What is released—result-level data and offline analysis scripts—establishes that the numbers can be recomputed. It cannot establish where the idea came from.

We note a further tension internal to the released material. The staged reports characterise the contribution in measured terms: phase-stepping interferometry is "a mature technique", and "the methodological novelty is not the demodulation itself but its application to extract a computationally meaningful bilinear observable". The abstract of the manuscript describes the same object as a "previously unreported physical mechanism".

9 Discussion

9.1 Limits of the computational interpretation

Scale and effective dimensionality. The released computational demonstration contains eight fixed tokens and 64 ordered pairs, all derived from one deterministic random-phase codebook. It therefore does not test generalisation to a larger vocabulary, to newly encoded tokens, to another random codebook, or to an independently acquired physical operator. The target work itself identifies vocabulary scale as a limitation and notes, in the authors' released Report 4, that a related token-family representation previously exhibited degeneracy-boundary collapse at $N = 16$.

Nominal detector dimensionality should also be distinguished from effective dimensionality on the evaluated data. At the main-text bin size of 25 pixels, we reproduce a participation-ratio effective rank of 3.00 for the released Complex-B feature ensemble, despite hundreds of nominal detector channels; at bin size 10 the corresponding value is 9.62. These quantities are properties of the observed eight-token feature ensemble, not universal rank bounds on the apparatus. They nevertheless show that the high-dimensional capacity invoked in the computational interpretation was not demonstrated on the benchmark used to support it: most observed variation is concentrated in a small number of effective complex directions, and near-saturated accuracy is obtained on a correspondingly small, repeatedly observed vocabulary.

Speed and energy are unmeasured outlooks. The experiment establishes that the complex cross term can be recovered on the physical platform; it does not establish a system-level speed or energy advantage. In the released acquisition protocol, each ordered input pair is measured through four sequential phase settings. The overall protocol also contains blank-control measurements, camera readout, calibration, normalisation, demodulation, and digital classification. Blank controls may be reused across multiple pairs and therefore should not automatically be counted as a per-pair cost, but they remain part of the demonstrated system.

No end-to-end latency, pair-score throughput, energy per recovered score, total system power, or amortised calibration cost is reported in the public materials. Nor is the demonstrated implementation compared under a matched accuracy and precision target with a GPU, FPGA, or digital complex correlator. The current experiment acquires the ordered pairs individually; it does not demonstrate parallel construction of an $N \times N$ score matrix, softmax normalisation, value projection, or value aggregation. Future spatial, wavelength, or temporal parallelisation may change this assessment, but it is not measured here. Statements concerning high-speed or energy-efficient Transformer-like hardware should therefore be read as prospective motivation rather than experimental findings of the present study.

Protocol evolution and confirmatory status. The released reports document an evolving analysis rather than one fixed confirmatory protocol. Report 4 and Report 5 describe different cross-split constructions and report materially different behaviour across feature families. The later analysis additionally explores detector bin sizes, output-channel budgets, sparse-tap selections, raw-input controls, random-operator controls, and revised task definitions. Such exploration is scientifically useful and, when reported transparently, is not itself an error. Because the documented Task-D protocols are not identical, we do not treat numerical differences between the two reports as controlled method-to-method comparisons.

It does, however, change the evidential status of the resulting performance comparisons. The public materials do not document when the token vocabulary, task definitions, split rules, bin size, output budget, or highlighted operating point were fixed relative to inspection of the corresponding outcomes. No independently acquired confirmation set or new physical operator is used after these choices. We therefore interpret the released performance landscape as exploratory. A confirmatory computational claim would require a frozen protocol evaluated on new tokens, a new codebook, an independent acquisition, or another physical realisation without further selection of readout configuration.

10 Scope and unaudited claims

This audit is deliberately claim-specific. We examine the released Result-4 pipeline because it provides the empirical basis for the manuscript's claims concerning optical bilinear computation, semantic pair representation, and the analogy to a core operation in Transformer attention. We reproduce and stress-test the public data, scripts, feature construction and evaluation protocol associated with that result.

We do not present a complete audit of the two preceding studies. In particular, we have not independently evaluated the optical acquisition efficiency or the focusing-enhancement deficit in the transmission-matrix reproduction (Result 2), nor have we reconstructed the full theory-to-measurement argument, mixed-state synthesis or uncertainty propagation in the majorization-order study (Result 3). We therefore draw no conclusion about the correctness, novelty or autonomy of those results. Their inclusion in the target manuscript does not, however, resolve the benchmark-integrity failures identified for Result 4, nor does it provide evidence for the specific computational claims tested here.

Our operator analysis is similarly bounded. We distinguish identities that follow from the ideal shared-operator model from properties of the released experimental estimator. The empirical deviations reported in Sec. 5 do not identify their physical cause. They may arise from unequal propagation operators in the two input arms, residual self-reference terms, imperfect phase stepping, calibration error, drift, or another unmodelled component. Resolving these alternatives requires measurements and hardware information not present in the public deposit.

Finally, statements about absent evidence refer only to the public record available to us as of 30 July 2026: arXiv:2604.27092v1 and the Zenodo deposit 10.5281/zenodo.19890402. The deposit states that the formal Supplementary Information submitted to the journal, the Qiushi Engine core system and the hardware-control software are not included. We therefore distinguish "not present in the public materials" from "not performed". In particular, we cannot determine whether an operator-matched reconstruction, additional agent-provenance records or alternative evaluation protocols exist in non-public submission materials.

11 What we do not claim

We find no evidence of fabrication. The raw frames, repeat structure, drift monitoring and transmission-matrix data are substantial and internally consistent, and the reported focusing and coherence-order results are plausible on their face. The automation demonstrated—sustained agent control of a real optical platform over hundreds of steps—is a genuine engineering result that our findings do not touch. Our claim is narrower and, we think, more serious: the specific evidence offered for the specific central conclusion does not support it.

12 Evidence needed to adjudicate the claims

The deficiencies identified above are resolvable. They require different evidence for the computational, physical, and autonomy claims; no single additional accuracy value would address all three.

Agent provenance. Attribution of an idea to an autonomous agent requires disclosure of the backbone model and exact version, inference settings, initial and system prompts, retrieval sources, progressively disclosed skills, candidate directions, direction-selection decisions, and human interventions. The complete time-ordered Meta-Trace and tool trajectory should be released in a form that distinguishes contemporaneous records from summaries produced after the experiment. If safety, licensing, or instrument-access constraints prevent release of particular implementation details, the omitted fields and their relevance to the claimed discovery should be identified explicitly. Result-level data alone cannot establish the causal origin of the hypothesis.

Prior-availability control. The relevant question is not whether a current language model can reproduce the mechanism after publication, but whether the backbone available to the agent before the study already contained or could readily elicit the same proposal. This should be tested using the exact original model snapshot, without the agent scaffold or experimental feedback, from the same initial platform description. The prompt, decoding parameters, number of independent queries, retrieval permissions, and a prospectively defined scoring rule should be fixed before inspection of the outputs. The assessment should separately record whether the model proposes the interference cross term and whether it identifies the underlying phase-stepping and coherent-detection principles as established knowledge. Such a control would measure prior availability; it would not by itself prove the causal origin of the observed trajectory.

Operator-matched physical control. For each detector pixel, the ideal shared-operator model predicts

$$B_p(x, y) = (TE_x)^*_p (TE_y)_p. \tag{2}$$

For a binned channel $\mathcal{B}_k$, the corresponding reconstruction is

$$\bar{B}_k^{\text{digital}}(x,y) = \sum_{p\in\mathcal{B}_k} (TE_x)_p^*(TE_y)_p. \tag{3}$$

The decisive control is therefore to measure the two propagated complex fields separately under the same physical operator and compare this digital reconstruction, channel by channel, with the jointly demodulated $\bar{B}_k^{\text{physical}}$. Agreement should be quantified in amplitude, phase, residual structure, repeatability, and downstream performance under identical feature budgets and splits. If the two input arms implement different operators, the corresponding $T_x^\dagger T_y$ model should be stated and tested instead. This experiment would determine whether joint measurement supplies information beyond separately measured complex fields; intensity products and random-operator pseudo-B controls do not answer that question.

Benchmark with a valid unit of generalisation. A confirmatory benchmark should contain relational structure in the inputs rather than attach semantic labels to independent random vectors. It should use tokens, codebooks, or physical inputs not involved in configuration selection. All repeated measurements of the same physical condition must remain in the same fold whenever the claim concerns generalisation to new conditions. Self-pairs should either be excluded from relation evaluation or reported as a separate stratum, and performance should be compared with measurement-free structural rules and an exact or permutation null over admissible label assignments. Pair identity may be evaluated as repeat recognition, but it should not be interpreted as out-of-pair generalisation. The final test should use a new acquisition session, codebook, vocabulary, or physical operator after the evaluation protocol has been frozen.

Independent agent and hardware replications. A single successful trajectory cannot establish that the reported outcome is a reproducible property of the agent architecture. The open-ended study should be repeated independently from the same initial information, with all runs and unsuccessful directions retained. At least three complete runs would provide a minimal existence check, although more are required to estimate a meaningful discovery rate for a stochastic system. Replication should also separate agent stochasticity from hardware specificity by including, where feasible, an independent acquisition session or physical realisation. The reported outcomes should include the frequency with which the mechanism is proposed, selected, experimentally supported, rejected, or replaced by another direction.

Pre-registered confirmation. Before the confirmatory run, the token set, task definitions, train–test unit, treatment of self-pairs, detector binning, output budget, feature families, classifier, hyperparameters, primary endpoint, and statistical comparison should be frozen in a time-stamped protocol. Exploratory searches over bin size, n_{out}, task construction, or feature family may be reported in full, but the selected configuration should then be evaluated on untouched data without further adaptation. This separation would turn the released performance landscape from an exploratory result into a test of a specified computational claim.

Together, these additions would not require treating autonomous discovery as a binary label. They would allow readers to distinguish four progressively stronger conclusions: automated execution of a known protocol, recombination of available scientific knowledge, generation and testing of a new engineering interpretation, and experimental identification of a finding not available to the model or its operators beforehand.

Data and code availability

All analysis code, the audit modifications and the full result files are at `https://github.com/[user]/discovery-audit`. The audit consumes only the authors' public deposit (DOI

10.5281/zenodo.19890402); no data of our own were generated, and the deposit is not redistributed. Every file read from the deposit is recorded with its SHA-256 in `results/input_manifest.json`; the roll-up digest over all 1543 consumed files is `b273f7f6c8d5bd8d7bb849e1d9908ea35a8ec171b9870b7cd2cb27b4963513` so any subsequent change to the deposit is detectable. The environment is pinned in `environment.yml` and `requirements-lock.txt`.

Correspondence with the authors

On [date] we sent the corresponding authors a set of technical questions covering the operator-matched control, the scoring protocol, the timing of the bin-size selection, and the backbone model. [Their response is incorporated above / No response had been received by [date].]

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